CS-UY 1134 Homework 2
pd2752

Question 1

(a) Claim: $5n! + 2n^2 + 3n$ is $O(n!)$.
For $n \geq 4$, $n! \geq n^2$ and $2n^2 + 3n \leq 3n^2$. Therefore,
$5n! + 2n^2 + 3n \leq 5n! + 3n! = 8n!$.
Taking $c = 8$ and $n_0 = 4$ satisfies the definition of $O(n!)$.

(b) Claim: $\sqrt{7n^2 + 2n - 8}$ is $\Theta(n)$.
For $n \geq 2$,
$4n^2 \leq 7n^2 + 2n - 8 \leq 9n^2$.
Taking square roots gives
$2n \leq \sqrt{7n^2 + 2n - 8} \leq 3n$.
Thus $c_1 = 2$, $c_2 = 3$, and $n_0 = 2$ satisfy the definition of $\Theta(n)$.

(c) Since $d(n) = O(f(n))$, there are constants $c_1 > 0$ and $n_1$ such that $0 \leq d(n) \leq c_1 f(n)$ for $n \geq n_1$. Since $e(n) = O(g(n))$, there are constants $c_2 > 0$ and $n_2$ such that $0 \leq e(n) \leq c_2 g(n)$ for $n \geq n_2$. For $n \geq \max(n_1, n_2)$, multiplying the two inequalities gives $d(n)e(n) \leq c_1c_2 f(n)g(n)$.
Therefore $d(n)e(n) = O(f(n)g(n))$, using $c = c_1c_2$.

Question 2

example1: $\Theta(n^2)$. The inner loop executes $j + 1$ times for each $j$ from 0 through $n - 1$. The total is $\sum(j + 1) = n(n + 1)/2$.

example2: $\Theta(n)$. Its one loop has $n$ iterations, and each iteration does a constant number of operations.

example3: $\Theta(\log n)$. After $t$ iterations, $i = 2^t$. The loop stops when $2^t \geq n^2$, so $t = \lceil \log_2(n^2) \rceil = \Theta(\log n)$.

example4: $\Theta(n)$. The outer loop uses $i$ values $n$, $\lfloor n/2 \rfloor$, $\lfloor n/4 \rfloor$, and so on. The total inner-loop work is bounded above by $n + n/2 + n/4 + \ldots < 2n$ and is at least $n$ from the first iteration. It is therefore $\Theta(n)$.